

Team LONG PULL — 2026-07-08. α is RULED (banked). Arrange around the FOUR FLAVOR MATRICES — one construction, ~17 of the 26. The Shilov spacing (diagonal/masses) is the single bottleneck; the overlaps (off-diagonal/mixing) are forward now. date + SOD.

α RULED (Casey) $\rightarrow$ banked

Casey ruled $\alpha \rightarrow$ derived-strong: discrete $137 = N_c^3 \cdot n_C + \text{rank (forced)} + \text{continuous Wyler (0.6 ppm, forward)}$ — the two-sided ellipse. K659 closed.

FIRST — reconcile the count (Cal + Keeper, load-bearing; we're citing numbers)

Make the currency honest before we quote a count: - α : partial $\rightarrow$ RULED (derived-strong). $\Delta m_{31}^2 / \Delta m_{21}^2 = 33$: banked yesterday (Chern, T1972, 0.7σ). - m_d/e: the ledger lists it banked; the scoreboard scores a 67% MISS — REMOVE from the banked list (or replace the form). This is the known ledger error. - Deliverable (Cal): one reconciled derived-strong list (forced-mechanism + $<\sigma$ + not-cheap), a single honest number. Running tally says ~10; the honest number is whatever survives m_d/e coming off. Cite the reconciled number, not the tally.

THE ARRANGEMENT — the four flavor matrices (the simplification)

Each sector = ONE matrix on its SO(3) generation triplet: $> M = [\text{diagonal: bare deposits} = \text{Shilov formal-degree spacing}] + [\text{off-diagonal: Bergman overlaps}]$. $>$ Diagonalize $\rightarrow$ eigenvalues = masses, eigenvectors = mixing. Four matrices (down, up, charged-lepton, neutrino) = ~17 of the 26 (9 masses + 4 CKM + 4 PMNS), from ONE construction. - OFF-DIAGONAL (mixing) = forward NOW (the overlap geometry; Grace's $r_2=1/2 \rightarrow$ Cabibbo). Bank as the localizations land — NO Wolfenstein-power fits (Lyra F490: $V_{cb}=\lambda^2$, $V_{ub}=\lambda^3$ are 20%/factor-3 misses, reverse-read). - DIAGONAL (masses) = gated on the Shilov spacing — the one bottleneck below.

THE BOTTLENECK — the Shilov formal-degree spacing (GRACE) + a sharpening that may crack it cheaply

Masses = the eigenvalue SPACING (dimensionless RATIOS; the 900 $\times$ is a ratio, NOT a scale — the anchor is time, Keeper's theorem). Grace's bolt was "reference-gated Plancherel (Γ -products, genus-flip trap)." Try the ALREADY-DERIVED formal degree FIRST — it's free and target-innocent: - Elie already derived $d(\nu) = (5/2-\nu)(1-\nu)(2-\nu)(3-\nu)(4-\nu)$ — the Harish-Chandra formal degree over the 5 noncompact roots of $B_3 = \text{SO}(5,2)_C$. Target-innocent (root-system data), no reference, predates the mass application. - The three generations sit at three ν -strata (KW positions, also predate). Evaluate $d(\nu)$ at the three strata $\rightarrow$ does the SPACING give the mass ratios ($\{1, 20, 900\}$ down; the lepton/up/ ν ratios)? - If yes $\rightarrow$ the masses bank FORWARD (the counts *derived* from $d(\nu)$, not reverse-read — Grace's held down-ladder leaf closes, and the diagonal of all four matrices is done). This is the highest-value forward item on the board. Only if $d(\nu)$ -at-strata fails do we escalate to the reference-gated Plancherel weight (the genus-flip trap). Try the free path first.

Per-CI (independent lanes, no dog-pile)

- GRACE: the Shilov spacing — $d(\nu)$ at the three ν -strata first (target-innocent). If the spacing gives the mass ratios, the diagonal banks across sectors. Then α root-lock #2 (does the discrete count 137 fall out of the $S^2 \times S^1$ coupling *curvature*? — confirms the trunk).
- LYRA: the overlaps (off-diagonal/mixing) — force the localizations $r_2=1/2$, $r_3 \rightarrow V_{us}, V_{cb}, V_{ub}$ as direct overlaps (no power-law fits); the PMNS overlaps. Hand Keeper each form as it lands.
- ELIE: build + diagonalize the four matrices (down, up, lepton, ν); force the ν mass coefficients $\{7/12, 10/3\}$ as boundary invariants; the up-sector (m_u, m_c, m_t). Report whether the held leaves θ_{23}, θ_{12} fall out of the eigenvectors (forced) or must be inserted (held).
- KEEPER: adjudicate each leaf (forced count + understood residual, K665 bar); reconcile the ledger with Cal; update the tree Artifact (root-lock #1 proven, α banked); over-sell #7 watch.
- CAL: the ledger/scoreboard reconciliation (the honest count) + post-landing cold-reads.

The held leaves — BUILD the matrices, don't argue

$\theta_{23} = 4/7$ and θ_{12} were HELD (value-form/form-cheap). The four-matrix construction decides them: if diagonalization gives θ_{23} , θ_{12} from the geometry (eigenvectors), they're forced $\rightarrow$ bank. If they must be inserted, they stay held. Build, harvest, tier — the construction is the arbiter, not the tidiness.

Disciplines armed

Forced count (Shilov spacing = $d(\nu)$ -at-strata, target-innocent — NOT reverse-read). Overlaps forward (no Wolfenstein-power fits). Discrete/continuous (count forced + correction continuous). Ellipse (drops out) / epicycle (inserted) / floor (irreducible). Over-sell #7 watch — six caught this run; the four-matrix frame is powerful enough to bank illusions if the entries aren't real integrals (Elie's guard: entries = real Bergman/formal-degree data, never inserted). Cite the reconciled count, not the tally. Cal is a later reviewer. Count honest per leaf.

What could bank (realistic, IF forward)

- Masses (down, and by the same $d(\nu)$ the lepton/up/ ν diagonals) — IF $d(\nu)$ -at-strata gives the spacing. The big one: potentially several mass leaves at once, forward.
- Mixing (CKM/PMNS) — as the overlap localizations land (V_{us} the nearest).
- θ_{23} , θ_{12} — bank IFF they fall out of the eigenvectors.
- Honest: no cluster-banking (yesterday's lesson) — each leaf on its own mechanism.

Reserved for Casey

Mass-frame ratification (needed as masses bank) · the paper strategy (gated on the 26).

Long-pull target

The Shilov spacing tried forward via $d(\nu)$ -at-strata — if it lands, the flavor-matrix diagonals bank across sectors (the real prize). The mixing overlaps computed forward (V_{us} nearest). The four flavor matrices built + diagonalized (the ~ 17 -leaf simplification tested). The ledger reconciled to one honest count. α root-lock #2 confirms the trunk. Forward leaves bank; held leaves stay held; nothing over-sold.

— Keeper, 2026-07-08 long pull. α RULED (banked, discrete+continuous). Arrange around the FOUR FLAVOR MATRICES (diagonal = Shilov spacing/masses, off-diagonal = overlaps/mixing; ~ 17 of 26, one construction). Bottleneck = the Shilov spacing — SHARPENING: try Elie's already-derived $d(\nu) = (5/2 - \nu)(1 - \nu)(2 - \nu)(3 - \nu)(4 - \nu)$ at the three ν -strata FIRST (free, target-innocent) before the reference-gated Plancherel. Overlaps forward now (no Wolfenstein fits). Reconcile the count ($m_{d/e}$ MISS off the list). Held leaves decided by building, not arguing. Over-sell #7 watch. Count: $\alpha + \Delta m^2$ -ratio the two new; Cal reconciles the honest number.

ADD (Casey: count as high as we can JUSTIFY, via “what does it mean”) — the count-justifying lanes

Count-tracking (Keeper draft for Cal): honest firm derived-strong = 5 (θ_{QCD} , α [ruled], $\sin^2\theta_{13}$, Δm^2 -ratio, $m_{t/\nu}$). Running “10” included the $m_{d/e}$ MISS (removed) + value-forms (held). Two JUSTIFIABLE upgrade paths, both generative (investigate what it is, don't assume):

- ELIE — the QED-continuum check (justifies the lepton-ratio upgrades): $m_{\mu/e} = (24/\pi^2)^6$ and $m_{\tau/e} = 49.71$ scored “APPROX/high- σ ” only because a rational can't equal a continuum. The COUNT is forced ($24 = N_c \cdot |W(B_2)|$, forced). TEST: does the QED radiative correction to the pole-mass RATIO account for the $\sim 10^{-4}$ (m_{μ}) / $\sim 5 \times 10^{-4}$ (m_{τ}) residual — magnitude AND sign? YES $\rightarrow$ forced count + continuum $\rightarrow$ DERIVED, +1 to +2. NO $\rightarrow$ APPROX, held. Standard QFT, not a fudge. (See Keeper_lepton_ratio_upgrade_investigation.)
- CAL — resolve $\sin^2\theta_{12}$ (scoreboard FORCED-matrix-element vs Elie form-cheap) $\rightarrow \pm 1$.

Justifiable reconciliation ceiling today: $5 \rightarrow \sim 7-8$ — EARNED by the QED test + the $\sin^2\theta_{12}$ call, not asserted. PLUS whatever the four flavor matrices bank forward (mixing overlaps, mass spacing).

Discipline: this is “investigate $\rightarrow$ what does the residual/form MEAN,” NOT “hunt for a justification.” The QED test can REFUSE the upgrade (wrong sign/size = geometric, held). We bank what the computation forces, not what we’d like.